

CHE 201 Spring 2020 Final Exam

Thanh Nguyen

TOTAL POINTS

100 / 100

QUESTION 1

1 15 pts

1.1 a 12 / 12

✓ - **0 pts** Correct

- **2 pts** first is incorrect
- **2 pts** second is incorrect
- **2 pts** third is incorrect
- **2 pts** fourth is incorrect
- **2 pts** fifth is incorrect
- **2 pts** sixth is incorrect

1.2 b 3 / 3

✓ - **0 pts** Correct

- **3 pts** Not done or wrong
- **2 pts** not explained

QUESTION 2

2 10 pts

2.1 a 5 / 5

✓ - **0 pts** Correct

- **1 pts** calc error
- **3 pts** h₂s is wrong
- **2 pts** formula is wrong
- **2 pts** explanation is not clear
- **5 pts** not done

2.2 b 5 / 5

✓ - **0 pts** Correct

- **2 pts** pr(T₁) is wrong
- **1 pts** calculation error
- **5 pts** not done or wrong
- **3 pts** formula is wrong

QUESTION 3

3 10 pts

3.1 a 2 / 2

✓ - **0 pts** Correct

- **2 pts** Not done or wrong

3.2 b 2 / 2

✓ - **0 pts** Correct

- **2 pts** wrong or not done
- **1 pts** one is wrong

3.3 c 6 / 6

✓ - **0 pts** Correct

- **6 pts** not done or wrong
- **2 pts** u value is wrong
- **2 pts** s value is wrong

QUESTION 4

4 32 / 32

✓ - **0 pts** Correct

- **2 pts** sketch is not drawn or incorrect
- **3 pts** engineering model is not listed
- **4 pts** simplified energy balance is not done
- **4 pts** work is not calculated or wrong
- **4 pts** heat is not calculated or wrong
- **5 pts** entropy change is not calculated
- **5 pts** sigma is not calculated
- **2 pts** pv diagram is wrong
- **3 pts** Ts diagram is wrong
- **2 pts** v₂ is wrong
- **2 pts** u₂ is wrong
- **2 pts** calculation error
- **2 pts** s₂ is wrong
- **3 pts** did not multiplied with mass
- **5 pts** v₁, u₁, s₁ are wrong
- **2 pts** T in sigma calc is wrong

QUESTION 5

5 5 33 / 33

✓ - 0 pts Correct

- 3 pts sketch is not drawn
- 3 pts engineering model is not listed
- 9 pts mass flow rate of inlet not calculated or

wrong

- 9 pts mass flow rate of outlet is not calculated
- 9 pts entropy production is not calculated
- 2 pts s_3 is wrong
- 2 pts h_3 is wrong
- 2 pts h_2 is wrong
- 2 pts s_2 is wrong

1.)

A. True or False

1. False

2. False

3. True

4. True

5. False

6. True

B. Fill in the blank

less than

$$\Delta S_{AB} = \int_A^B \left(\frac{\delta Q}{T} \right)_b + \bar{G}_{AB}$$

Since $\bar{G}_{AB} = 0$ (reversible) and $Q_{AB} = 0$ (adiabatic) $\rightarrow \Delta S_{AB} = 0$

$$\Delta S_{\text{cycle}} = \cancel{\Delta S_{AB}} + \Delta S_{BA} \rightarrow \Delta S_{BA} = 0$$

$$\Delta S_{BA} = \int_B^A \left(\frac{\delta Q}{T} \right)_b + \bar{G}_{BA} \rightarrow \int_B^A \left(\frac{\delta Q}{T} \right)_b = -\bar{G}_{BA}$$

Since $\bar{G}_{BA} > 0$ (irreversible) $\rightarrow Q < 0$

1.1 a 12 / 12

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- 2 pts second is incorrect
- 2 pts third is incorrect
- 2 pts fourth is incorrect
- 2 pts fifth is incorrect
- 2 pts sixth is incorrect

1.)

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2. False

3. True

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1.2 b 3 / 3

✓ - **0 pts** Correct

- **3 pts** Not done or wrong

- **2 pts** not explained

2.)

a) $\frac{\dot{W}_{cv}}{\dot{m}} = 150 \text{ kJ/kg}$

Using table A-4, at $p_1 = 10 \text{ bar}$

• $T_a = 280^\circ\text{C} \rightarrow h_a = 3008.2 \text{ kJ/kg}; s_a = 7.0465 \text{ kJ/kg}\cdot\text{K}$

• $T_b = 320^\circ\text{C} \rightarrow h_b = 3093.9 \text{ kJ/kg}; s_b = 7.1962 \text{ kJ/kg}\cdot\text{K}$

Using interpolation: $\frac{T_1 - T_a}{T_b - T_a} = \frac{h_1 - h_a}{h_b - h_a} = \frac{s_1 - s_a}{s_b - s_a}$

$\rightarrow h_1 = 3051.05 \text{ kJ/kg}$ and $s_1 = 7.12135 \text{ kJ/kg}\cdot\text{K}$

$\rightarrow s_{2s} = s_1 = 7.12135 \text{ kJ/kg}\cdot\text{K}$

Using table A-4, at $p_2 = 5 \text{ bar}$

• $s_x = 7.0592 \text{ kJ/kg}\cdot\text{K} \rightarrow h_x = 2855.4 \text{ kJ/kg}$ (at $T_x = 200^\circ\text{C}$)

• $s_y = 7.2307 \text{ kJ/kg}\cdot\text{K} \rightarrow h_y = 2939.9 \text{ kJ/kg}$ (at $T_y = 240^\circ\text{C}$)

Using interpolation: $\frac{s_{2s} - s_x}{s_y - s_x} = \frac{h_{2s} - h_x}{h_y - h_x} \rightarrow h_{2s} = 2886.02 \text{ kJ/kg}$

$\eta = \frac{(\dot{W}_{cv}/\dot{m})}{h_1 - h_{2s}} = \frac{150 \text{ kJ/kg}}{(3051.05 - 2886.02) \text{ kJ/kg}} = 0.9065$

b) Since $s_2 = s_1 \rightarrow \frac{Pr(T_2)}{Pr(T_1)} = \frac{P_2}{P_1}$

Using table A-22, at $T_1 = 650 \text{ K} \rightarrow Pr(650 \text{ K}) = 21.86$

$\rightarrow Pr(T_2) = Pr(T_1) \frac{P_2}{P_1} = (21.86 \text{ bar}) \frac{(2 \text{ bar})}{(10 \text{ bar})} = 4.372$

Using table A-22:

• $T_a = 410 \text{ K}, Pr(T_a) = 4.153$

• $T_b = 420 \text{ K}, Pr(T_b) = 4.522$

Using interpolation: $\frac{T_2 - T_a}{T_b - T_a} = \frac{Pr(T_2) - Pr(T_a)}{Pr(T_b) - Pr(T_a)} = \frac{4.372 - 4.153}{4.522 - 4.153}$

$\rightarrow T_2 = 415.93 \text{ K}$

2.1 a 5 / 5

✓ - 0 pts Correct

- 1 pts calc error
- 3 pts h2s is wrong
- 2 pts formula is wrong
- 2 pts explanation is not clear
- 5 pts not done

2.)

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$\rightarrow T_2 = 415.93 \text{ K}$

2.2 b 5 / 5

✓ - 0 pts Correct

- 2 pts $\text{pr}(T1)$ is wrong
- 1 pts calculation error
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3.)

a) $g(T, p) : dg = -s dT + v dp$

$$\rightarrow \left(\frac{-\delta s}{\delta p} \right)_T = \left(\frac{\delta v}{\delta T} \right)_p$$

b) $dh = T ds + v dp$

$$\frac{\delta h}{\delta s \delta p} = \left(\frac{\delta T}{\delta p} \right)_s ; \quad \frac{\delta h}{\delta p \delta s} = \left(\frac{\delta v}{\delta s} \right)_p$$

c) $\psi = u - Ts$

Using table A-3 water in saturated liquid at $p_{\text{sat}} = 10 \text{ bar}$ and $T_{\text{sat}} = 179.9^\circ\text{C}$
 $\rightarrow$ water in compressible liquid

Using table A-2:

$$\rightarrow u(150^\circ\text{C}) \approx u_f(150^\circ\text{C}) = 631.68 \text{ kJ/kg}$$

$$\text{and } s(150^\circ\text{C}) \approx s_f(150^\circ\text{C}) = 1.8418 \text{ kJ/kg}\cdot\text{K}$$

$$T = 150^\circ\text{C} + 273 = 423 \text{ K}$$

$$\psi = u(150^\circ\text{C}) - Ts(150^\circ\text{C})$$

$$= (631.68 \text{ kJ/kg}) - (423 \text{ K})(1.8418 \text{ kJ/kg}\cdot\text{K})$$

$$= -147.40 \text{ kJ/kg}$$

$$\Psi = m\psi = (3 \text{ kg})(-147.40 \text{ kJ/kg}) = -442.20 \text{ kJ}$$

3.1 a 2 / 2

✓ - 0 pts Correct

- 2 pts Not done or wrong

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3.2 b 2 / 2

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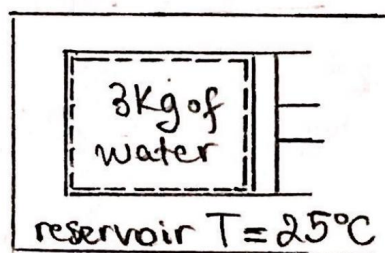
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3.3 C 6 / 6

✓ - 0 pts Correct

- 6 pts not done or wrong
- 2 pts u value is wrong
- 2 pts s value is wrong

4.) Sketch



State 1

- $T_1 = 120^\circ\text{C}$
- $p_1 = 3\text{ bar}$

State 2

- $x = 0.6$

Engineering Model

1. Closed system
2. $Q \neq 0$
3. $\Delta KE = \Delta PE = 0$
4. Constant pressure $\Delta p = 0$
5. Work is only in the form of ΔV

Energy balance

$$\Delta E = \Delta U + \cancel{\Delta KE} + \cancel{\Delta PE} = Q - W \rightarrow \Delta U = Q - W$$

Entropy balance

$$s_2 - s_1 = \Delta S = \int_1^2 \left(\frac{\delta Q}{T} \right)_b + \sigma$$

Analysis

$$p_2 = p_1 = 3\text{ bar}$$

Water is in saturated liquid at $p = 3\text{ bar}$ and $T = 133.6^\circ\text{C}$

→ State 1: Water is in compressible liquid

Using table A-2:

$$\bullet v_1 \approx v_f(120^\circ\text{C}) = 1.0603 \times 10^{-3} \text{ m}^3/\text{kg}$$

$$\bullet u_1 \approx u_f(120^\circ\text{C}) = 503.50 \text{ kJ/kg}$$

$$\bullet s_1 \approx s_f(120^\circ\text{C}) = 1.5276 \text{ kJ/kg}\cdot\text{K}$$

Using table A-3, at $p_2 = 3\text{ bar}$ and $T_{\text{sat}} = 133.6^\circ\text{C}$

$$\bullet v_f = 1.0732 \times 10^{-3} \text{ m}^3/\text{kg}; v_g = 0.6058 \text{ m}^3/\text{kg}$$

$$\bullet u_f = 561.15 \text{ kJ/kg}; u_g = 2543.6 \text{ kJ/kg}$$

$$\bullet s_f = 1.6718 \text{ kJ/kg}\cdot\text{K}; s_g = 6.9819 \text{ kJ/kg}\cdot\text{K}$$

$$\rightarrow v_2 = (1-x)v_f + xv_g = (1-0.6)(1.0732 \times 10^{-3} \text{ m}^3/\text{kg}) + (0.6)(0.6058 \text{ m}^3/\text{kg})$$

$$= 0.3639 \text{ m}^3/\text{kg}$$

$$\rightarrow W = m \int_{v_1}^{v_2} p dv = mp(v_2 - v_1) = (3\text{ kg}) \times$$

$$= (3\text{ kg})(3\text{ bar}) \times \frac{10^5 \text{ N/m}^2}{1\text{ bar}} \times (0.3639 - 1.0603 \times 10^{-3}) \text{ m}^3/\text{kg} \times \frac{1\text{ kJ}}{1000\text{ J}}$$

$$= 326.56 \text{ kJ}$$

4.) (continue)

$$u_2 = (1-x)u_f + xu_g = (1-0.6)(561.15 \text{ kJ/kg}) + (0.6)(2543.6 \text{ kJ/kg})$$

$$= 1750.62 \text{ kJ/kg}$$

$$\Delta U = m\Delta u = m(u_2 - u_1) = (3 \text{ kg})(1750.62 - 503.50) \text{ kJ/kg}$$

$$= 3741.36 \text{ kJ}$$

$$\rightarrow Q = \Delta U + W = 3741.36 \text{ kJ} + 326.56 \text{ kJ} = 4067.92 \text{ kJ}$$

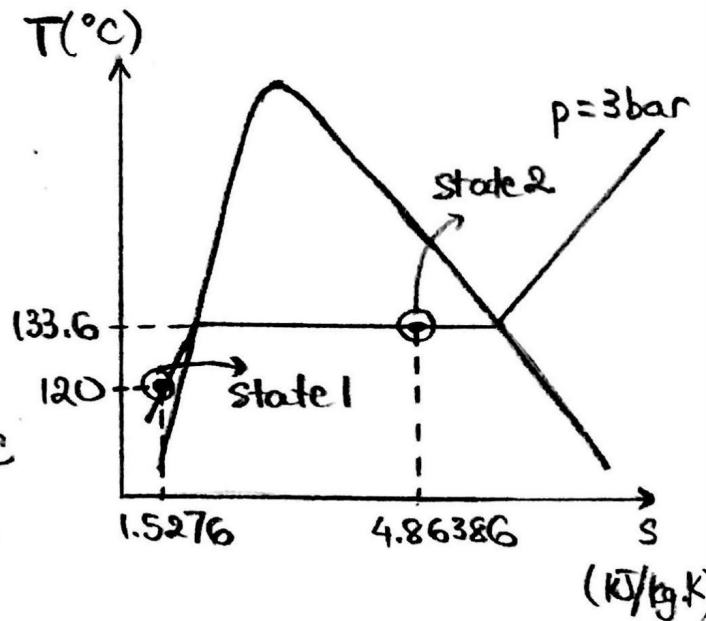
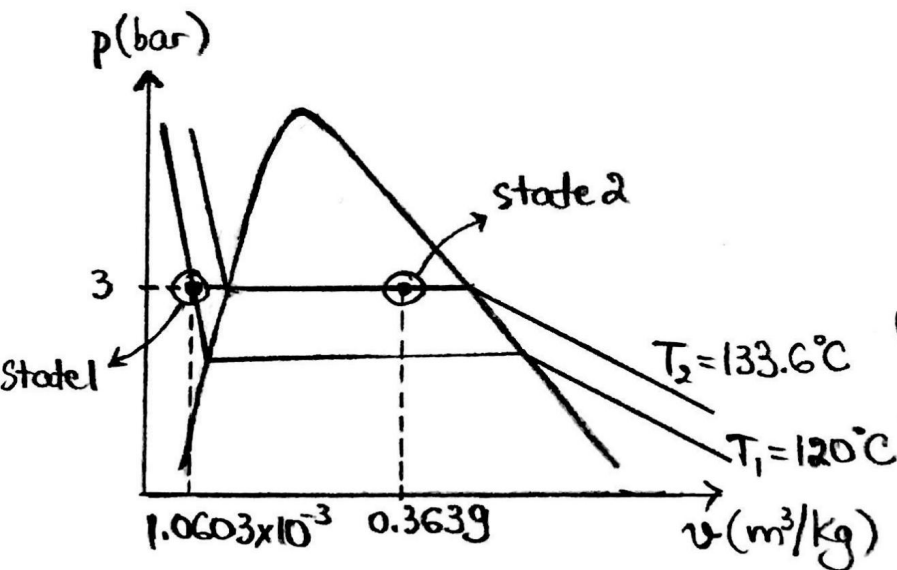
$$s_2 = (1-x)s_f + xs_g = (1-0.6)(1.6718 \text{ kJ/kg}\cdot\text{K}) + (0.6)(6.9919 \text{ kJ/kg}\cdot\text{K})$$

$$= 4.86386 \text{ kJ/kg}\cdot\text{K}$$

$$\Delta S = m\Delta s = (3 \text{ kg})(4.86386 - 1.5276) \text{ kJ/kg}\cdot\text{K} = 10.009 \text{ kJ/K}$$

$$\sigma = \Delta S - \int_1^2 \left(\frac{\delta Q}{T} \right)_b = 10.009 \text{ kJ/K} - \frac{3741.36 \text{ kJ}}{25^\circ\text{C} + 273} = -2.5459 \text{ kJ/K}$$

$\sigma < 0 \rightarrow$ impossible

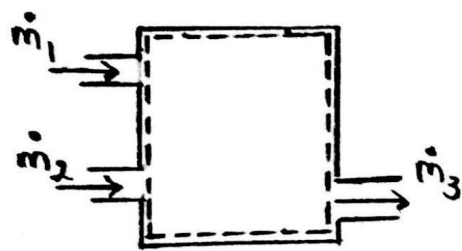


4 4 32 / 32

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- 2 pts s2 is wrong
- 3 pts did not multiplied with mass
- 5 pts v1,u1,s1 are wrong
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5.) Sketch



Stream 3

- saturated vapor
- $p_3 = 1 \text{ bar}$

Stream 1 (superheated steam)

- $p_1 = 5 \text{ bar}$
- $T_1 = 280^\circ\text{C}$
- $\dot{m}_1 = 20000 \text{ kg/min}$

Stream 2 (liquid)

- $p_2 = 3 \text{ bar}$
- $T_2 = 20^\circ\text{C}$

Engineering Model

1. Control volume
2. Steady state with 2 inlets and 1 outlet
3. $\Delta KE = \Delta PE = 0$
4. Adiabatic $\dot{Q}_{cv} = 0$
5. $\dot{W}_{cv} = 0$

Mass flow balance: $\frac{dm_{cv}}{dt} = 0 \rightarrow \sum \dot{m}_i - \sum \dot{m}_e = 0$
 $\rightarrow \dot{m}_1 + \dot{m}_2 - \dot{m}_3 = 0 \quad (1)$

Energy rate balance

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \sum \dot{m}_i \left[h_i + \frac{V_i^2}{2} + g z_i \right] - \sum \dot{m}_e \left[h_e + \frac{V_e^2}{2} + g z_e \right] = 0$$

$$\rightarrow \sum \dot{m}_i h_i - \sum \dot{m}_e h_e = 0 \rightarrow \dot{m}_1 h_1 + \dot{m}_2 h_2 - \dot{m}_3 h_3 = 0 \quad (2)$$

Entropy rate balance

$$\frac{ds_{cv}}{dt} = \sum \left(\frac{\dot{Q}_j}{T_j} \right)_b + \sum \dot{m}_i s_i - \sum \dot{m}_e s_e + \dot{\sigma}_{cv}$$

$$\rightarrow \dot{m}_1 s_1 + \dot{m}_2 s_2 - \dot{m}_3 s_3 + \dot{\sigma}_{cv} = 0 \quad (3)$$

Analysis

$$\dot{m}_1 = (20000 \text{ kg/min}) \times \frac{1 \text{ min}}{60 \text{ s}} = 333.3 \text{ kg/s}$$

From (1) $\rightarrow \dot{m}_3 = \dot{m}_1 + \dot{m}_2 \quad (4)$

Plug (4) into (2) $\rightarrow \dot{m}_1 h_1 + \dot{m}_2 h_2 - (\dot{m}_1 + \dot{m}_2) h_3 = 0$
 $\rightarrow \dot{m}_1 (h_1 - h_3) + \dot{m}_2 (h_2 - h_3) = 0 \rightarrow \dot{m}_2 = \dot{m}_1 \frac{h_1 - h_3}{h_3 - h_2}$

Using table A-4 at $p_1 = 5 \text{ bar}$, $T_1 = 280^\circ\text{C}$

$$\rightarrow h_1 = 3022.9 \text{ kJ/kg and } s_1 = 7.3865 \text{ kJ/kg}\cdot\text{K}$$

Using table A-2 at $T_2 = 20^\circ\text{C} = T_{\text{sat}}$ (compressible liquid) and $P_{\text{sat}} =$

$$\rightarrow h_2 \approx h_f(20^\circ\text{C}) + v_f(20^\circ\text{C}) [p_2 - P_{\text{sat}}(20^\circ\text{C})] = 83.963 \text{ kJ/kg}$$

$$\text{and } s_2 \approx s_f(20^\circ\text{C}) = 0.2966 \text{ kJ/kg}\cdot\text{K}$$

Using table A-3 at $P_{\text{sat}} = p_3 = 1 \text{ bar}$

$$\rightarrow h_3 = h_g = 2675.5 \text{ kJ/kg and } s_3 = s_g = 7.3594 \text{ kJ/kg}\cdot\text{K}$$

Thanh Nguyen

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5.) (continue)

$$\rightarrow \dot{m}_2 = (333.3 \text{ kg/s}) \frac{(3022.9 \text{ kJ/kg}) - (2675.5 \text{ kJ/kg})}{(2675.5 \text{ kJ/kg}) - (83.96 \text{ kJ/kg})} = 44.679 \text{ kg/s}$$

$$\rightarrow \dot{m}_3 = \dot{m}_1 + \dot{m}_2 = 333.3 \text{ kg/s} + 44.679 \text{ kg/s} = 377.979 \text{ kg/s}$$

$$\begin{aligned} (3) \rightarrow -\dot{Q}_{cv} &= \dot{m}_1 s_1 + \dot{m}_2 s_2 - \dot{m}_3 s_3 \\ &= (333.3 \text{ kg/s})(7.3865 \text{ kJ/kg} \cdot \text{K}) + (44.679 \text{ kg/s})(0.2966 \text{ kJ/kg} \cdot \text{K}) - \\ &\quad - (377.979 \text{ kg/s})(7.3594 \text{ kJ/kg} \cdot \text{K}) \\ &= -306.53 \text{ kJ/K} \cdot \text{s} \end{aligned}$$

$$\rightarrow \dot{Q}_{cv} = 306.53 \text{ kJ/kg} \cdot \text{s}$$

5 5 33 / 33

✓ - 0 pts Correct

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- 9 pts mass flow rate of inlet not calculated or wrong
- 9 pts mass flow rate of outlet is not calculated
- 9 pts entropy production is not calculated
- 2 pts s_3 is wrong
- 2 pts h_3 is wrong
- 2 pts h_2 is wrong
- 2 pts s_2 is wrong